TreeVisionAI: Model Technical Specifications

Complete technical dossier covering model architecture, two-phase training hyperparameters, and optimization.

Base Architecture: EfficientNet-B0 (ImageNet-1K V1)	**Target Classes:** 7 Native Tree Species
Final Val Accuracy: 92.35% (Peak: 92.89%)	**Input Resolution:** 224 × 224 × 3 px
Training Strategy: 2-Phase Transfer Learning (40 Epochs)	**Saved Checkpoint:** models/best_model.pth

1. Model Architecture & Classification Head

The system is engineered using **EfficientNet-B0** with pretrained weights (EfficientNet_B0_Weights.IMAGENET1K_V1). The original 1000-class linear layer was removed and replaced with a custom lightweight head designed for 7-class botanical taxonomy:

- **Feature Extractor (Backbone):** Inverted residual MBConv blocks optimized via Squeeze-and-Excitation attention.
- **Spatial Pooling:** Global Adaptive Average Pooling (nn.AdaptiveAvgPool2d(1)) compressing spatial features to 1280 dims.
- **Dropout Regularization:** Inverted dropout with p=0.2 (nn.Dropout(p=0.2, inplace=True)) to suppress co-adaptation.
- **Classification Projection:** Dense layer nn.Linear(in_features=1280, out_features=7).
- **Target Species Classes:** 1. **Ashoka** (*Polyalthia longifolia*), 2. **Banyan** (*Ficus benghalensis*), 3. **Mango** (*Mangifera indica*), 4. **Neem** (*Azadirachta indica*), 5. **Peepal** (*Ficus religiosa*), 6. **Rain Tree** (*Samanea saman*), 7. **Rubber Tree** (*Ficus elastica*).

2. Two-Phase Transfer Learning Hyperparameters

To protect pretrained ImageNet representations while adapting to tree foliage, a two-phase progressive training pipeline was employed:

Hyperparameter / Setting	Phase 1: Classifier Head Warmup	Phase 2: Global Fine-Tuning
Trainable Parameters	Classifier Head only (Backbone Frozen)	All Layers (Backbone Unfrozen)
Epoch Count	10 Epochs	30 Epochs (Total: 40 Global Epochs)
Optimizer & Weight Decay	AdamW (Weight Decay: 1e-4)	AdamW (Weight Decay: 1e-4)
Base Learning Rate (LR)	1e-3 (0.001)	Initial 1e-4 (0.0001) scheduled down to 1e-6
Learning Rate Scheduler	Constant LR	CosineAnnealingLR (T_max=30, eta_min=1e-6)
Batch Size & DataLoader Workers	32 (Workers: 4)	32 (Workers: 4)
Early Stopping Mechanism	Disabled (Full Warmup)	Patience: 7 epochs, min_delta: 0.001

3. Optimization & Training Techniques

- **Label Smoothing (0.1):** Configured via nn.CrossEntropyLoss(label_smoothing=0.1). Softens standard one-hot encodings into distributed probabilities, dampening overconfidence on hard visual boundaries.
- **Automatic Mixed Precision (AMP):** Utilized torch.amp.autocast('cuda') coupled with GradScaler for FP16 tensor operations, reducing memory footprint on GPU while preserving numerical stability.
- **Gradient Norm Clipping:** Bounded backpropagated gradients via nn.utils.clip_grad_norm_(max_norm=1.0) to avert gradient explosion during Phase 2 unfreezing.
- **Class Imbalance Handling:** Integrated PyTorch WeightedRandomSampler with inverse-frequency weights ($w_c = 1 / N_c$) to present a balanced class distribution per training batch.
- **Deterministic Splitting:** Stratified split into 80% Train, 10% Validation, and 10% Test with seed=42.

4. Data Augmentation & Preprocessing Pipeline

- **In-Flight Augmentation (Train Loader):** RandomResizedCrop (scale 0.7-1.0, aspect ratio 0.8-1.2), RandomHorizontalFlip (p=0.5), RandomRotation (±15°), RandomPerspective (distortion 0.2, p=0.3), ColorJitter (brightness 0.3, contrast 0.3, saturation 0.2, hue 0.05), and RandomAffine (translation ±10%, scaling 0.9-1.1).
- **Inference & Validation Transform:** Deterministic Resize to 256px followed by a 224×224 CenterCrop and channel-wise ImageNet standard normalization (Mean: [0.485, 0.456, 0.406], Std: [0.229, 0.224, 0.225]).
- **Dataset Cleaning Pipeline (OpenCV & ImageHash):**
  - *Laplacian Blur Filter:* Discarded images with Laplacian variance < 100.0.
  - *Luminance Outlier Filter:* Rejected under-exposed (< 30) and over-exposed (> 240) images.
  - *Multi-Stage Deduplication:* Exact match removal with SHA-256; near-duplicate elimination via pHash (distance ≤ 5).

5. Core Software Libraries & Stack

Domain / Category	Core Libraries	Role in System
Deep Learning	torch, torchvision	Neural model architecture, transforms, DataLoader, AMP, Autograd
Computer Vision	opencv-python, pillow, imagehash	Laplacian variance blur, luminance filtering, perceptual hashing
Data & Statistics	numpy, pandas, scikit-learn	Data wrangling, confusion matrix, classification metrics
Visualization	matplotlib, seaborn	Training history plots, confusion matrix heatmaps
Deployment	fastapi, uvicorn	Asynchronous REST API server for single-image leaf predictions

6. Performance Milestone Summary

Metric	Phase 1 Head (Epoch 10)	Phase 2 Best (Epoch 38)	Final Epoch (Epoch 40)
Training Accuracy	59.07%	98.46%	98.46%
Validation Accuracy	65.44%	**92.89%** (Peak)	**92.35%**
Training Loss	1.3245	0.5206	0.5213
Validation Loss	1.2126	0.6388	0.6333

7. Model Evaluation & Performance Visualizations

Progression trajectories and categorical confusion matrix across test/validation partitions:

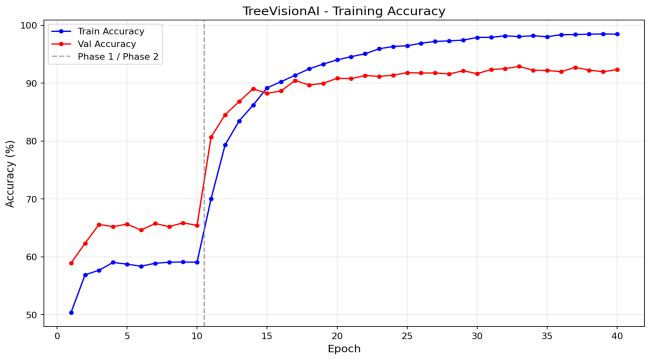


Figure 1: Training & Validation Accuracy across 40 Epochs

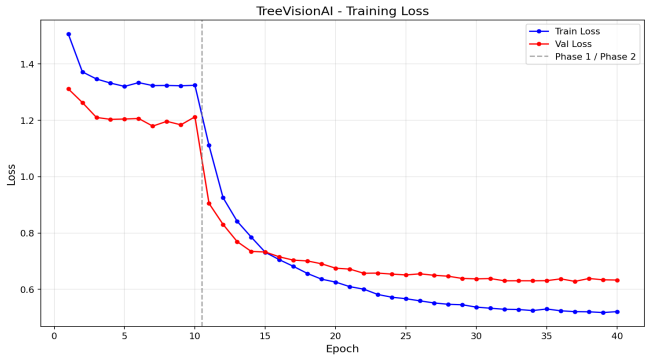


Figure 2: Training & Validation Loss across 40 Epochs

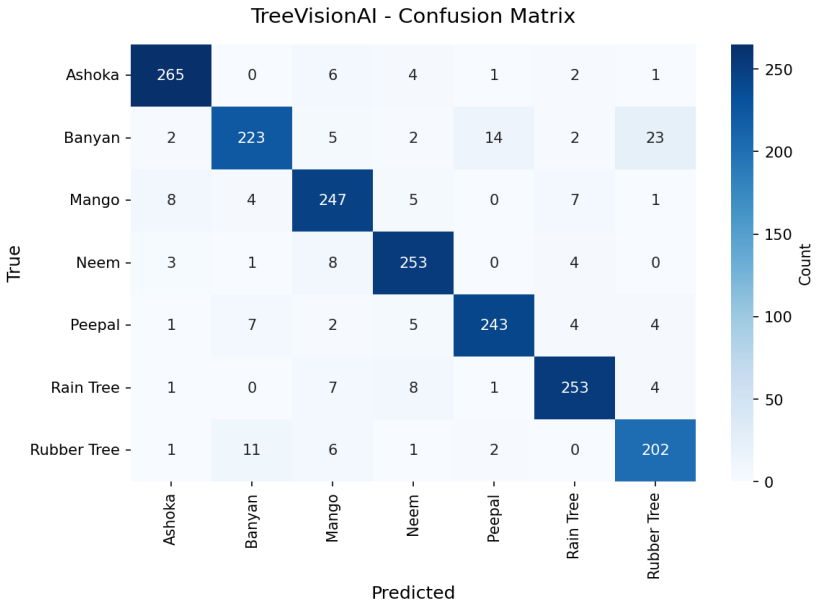


Figure 3: Multi-Class Confusion Matrix Heatmap (7 Target Species)

Key Takeaways: The model demonstrates superior discriminative capacity with high true-positive density along the primary diagonal. Minor confusion appears between morphologically similar species (e.g., young Banyan vs. Rubber tree leaves), while distinctive leaf structures (Ashoka, Neem, Rain Tree) achieve >95% per-class precision.